Mohamad Kazemi

Machine Learning for Sequential Data
Toronto, Ontario
Available Fall 2027

+1 (647) 381-6131
contact@mrkazemi.com
GitHub
LinkedIn
Google Scholar

Education

- **Master of Applied Science, Electrical Engineering & Computer Science** *2022–2024*
  York University Toronto, ON
  – GPA: 3.75 / 4.0.
  – Relevant coursework: Neural Networks and Deep Learning, Data Analytics and Visualization, Machine Learning Theory.

- **Bachelor of Science (Honours), Computer Science** *2018–2022*
  York University Toronto, ON
  – GPA: 3.97 / 4.0. First Class Standing with Distinction.
  – Relevant coursework: Machine Learning, Computer Vision, Probability, Linear Algebra, Algorithms, Operating Systems, Networks.

Publications & Manuscripts

- M. Kazemi and A. W. Eckford, *CHOIR: Channel Count, State Occupancy and Rates from Multichannel CFTR Recordings*, IEEE International Conference on Communications (ICC 2027), under review. PDF · Code.

Research Experience

- **MASc Thesis** *2022–2024*
  York University; Supervisor: Andrew W. Eckford Toronto, ON
  – **Thesis:** *Deciphering Ion Channel Dynamics: A Clustering Approach for Signal Analysis* (PDF). Asked whether a sequence model could recover hidden channel states from noisy patch-clamp recordings more reliably than clustering.
  – Built a two-stage pipeline: a neural network estimated the number of active channels, and an LSTM classified each channel's state from temporal context. Trained the pipeline on simulated recordings with realistic noise and tested it on real data.
  – It held up better than clustering baselines as noise and channel count increased, and classified single-channel recordings near-perfectly. The remaining errors clustered at state transitions.

- **Undergraduate Research Assistant (NSERC USRA)** *2022*
  York University; Supervisor: Hossein Kassiri Toronto, ON
  – Analyzed time-series brain recordings and built denoising pipelines for neural signals.

- **Undergraduate Research Assistant (NSERC USRA)** *2020*
  RISE Lab, York University; Supervisor: Pirathayini Srikantha Toronto, ON
  – Trained mixture-density networks (MDNs) and CNNs on power-grid simulations to detect anomalies and security breaches, reaching 95% accuracy.
  – Built Tableau and Matplotlib reports for stakeholders.

Awards & Fellowships

- **NSERC Undergraduate Student Research Award** *2020, 2022*
  Two-time competitive federal award funding full-time undergraduate research

- **Master's Graduate Fellowship** *2022–2024*
  York fellowship funding full-time MASc research York University

- **Continuing Student Scholarship** *2019–2021*
  York scholarship for sustained academic excellence York University

- **Dean's Honour Roll** *2022*
  Recognition for top academic standing Lassonde School of Engineering

Teaching, Mentoring & Service

- **Volunteer Instructor**, Excel Lassonde. Taught computer science and mathematics to students seeking additional help with coursework.
- **Research Mentor**, Molecular and Cellular Biology. Mentored a research assistant; cited for clarity and problem-solving.
- **Invited Panelist**, York University Faculty of Health, 2025. Led AI ethics cases on privacy, patient data integrity, and equity.
- **Student Representative, Faculty Council**, Lassonde School of Engineering. Reviewed academic-integrity cases and policy.
- **Director of Academic Affairs**, Lassonde Student Government. Built course-feedback channels and worked with department chairs.
- **Adjudicating Committee Member**, York University. Reviewed academic appeals and student-conduct cases.

Industry Experience

- **Senior Software Engineer** *May 2025–Nov 2026*
  Elation Health San Francisco, CA
  – Built LLM automation for clinical onboarding and legacy document ingestion using Anthropic Claude plugins and Prefect.
  – Built event-driven backend services on Kafka and Dapr for a multi-tenant EHR platform.
  – Added Redis caching for high-traffic paths in a multi-tenant deployment.

- **Senior Software Engineer** *Feb 2025–May 2025*
  Pine Toronto, ON
  – Built backend services for Pine Homes, a real-estate search platform with 40,000+ daily active users.
  – Added caching, rate limiting, and load balancing, and refactored legacy code into modular, monitored services.

- **Senior Full-Stack Developer** *May 2022–Feb 2025*
  Ourboro Toronto, ON
  – Built an in-house property price model that replaced a $30-per-query third-party service.
  – Shipped a generative AI assistant for customer support.